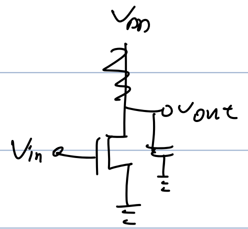
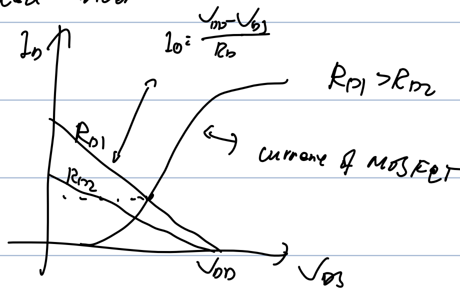
Inverter

Resistively loaded inverter

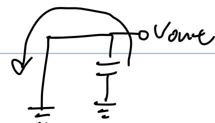




$R_D \uparrow \quad I_D \downarrow \quad P = I_D \cdot V_{DD} \downarrow$

$C = \frac{Q}{V} \Rightarrow C\frac{dV}{dt} = i_D \Rightarrow dt = C\cdot\frac{dV_{out}}{i_D}$

$t_{fall} = C\int_{0.9V_T}^{0.1V_T}\frac{dV_{out}}{i_D} = C\cdot\frac{0.8V_T}{\frac{1}{2}\beta(V_{2H}-V_t)^2} = \frac{1.6C\cdot V_{DD}}{\beta(V_{DD}-V_t)} \approx \frac{1.6CV_{DD}}{\beta}$



MOS sat

与 R_D 无关

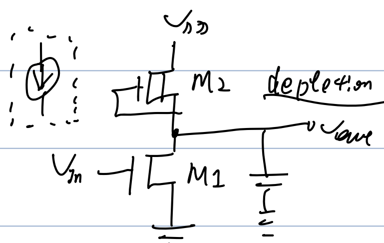
$t_{rise} = C\cdot\int_{0.1V_T}^{0.9V_T}\frac{dV_{out}}{i_D} = C\cdot\int_{0.1V_T}^{0.9V_T}\frac{dV_{out}}{(V_{DD}-V_{out})/R_D} = -CR_D\ln(V_{DD}-V_{out})\Big|_{0.1V_T}^{0.9V_T}$

$= CR_D\cdot\ln\frac{V_{DD}-0.1V_T}{V_{DD}-0.9V_T} = CR_D\ln 9 = \underline{2.2\,CR_D}$ 与 R_D 有关.

$R_D \uparrow \quad t_{rise} \uparrow$

NMOS loaded inverter

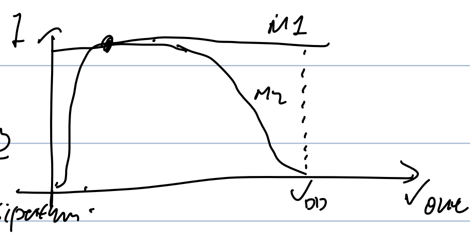
$\frac{\beta}{2}(V_{GS}-V_t)^2$



M_2 depletion mode.

$0 = V_{GS} > -|V_t|$

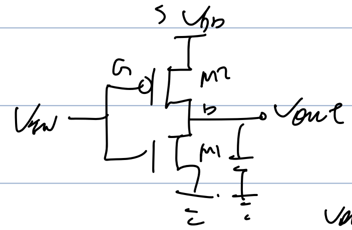
$V_{DS} > V_{GS} - (-|V_t|)$

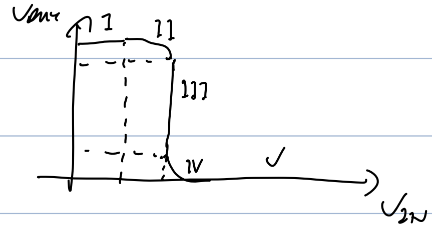


no R_D so faster switching

but still static power dissipation.

CMOS inverter.





Region	NMOS	PMOS
I	cut-off $V_{IN} < V_{tn}$	linear $V_{DD}-V_{IN} > \lvert V_{tp}\rvert$, $V_{DD}-V_{out} < V_{DD}-V_{IN}-\lvert V_{tp}\rvert$
II	saturation $V_{IN} \geq V_{tn}$, $V_{out} > V_{IN}-V_{tn}$	linear ...
III	...	...
IV	linear $V_{IN} \geq V_{tn}$, $V_{out} < V_{IN}-V_{tn}$	saturation $V_{DD}-V_{IN} > \lvert V_{tp}\rvert$, $V_{DD}-V_{out} > V_{DD}-V_{IN}-\lvert V_{tp}\rvert$
V	...	cut-off $V_{DD}-V_{IN} < \lvert V_{tp}\rvert$